Characterization of amylase activity under physicochemical conditions

DNS colorimetric assay · Commercial fungal and salivary amylase

pH 7

highest reported signal for both preparations

37 °C

highest reported signal in the temperature series

10 mM

source-labeled $CaCl_2$ response maximum

Executive summary

The supplied report contains one A540 reading per condition across pH, temperature, $CaCl_2$, starch concentration, and ethanol. Signal rises through the tested starch range and falls as source-labeled ethanol increases. These are descriptive endpoint patterns, not replicated or normalized enzyme-rate estimates.

Calibration refit

$A540 = 0.81833 \times C - 0.01578$; $R^2 = 0.99070$

No replicate readings or raw instrument export were supplied. Interpret results as preliminary descriptive evidence.

GRAPHICAL ABSTRACT

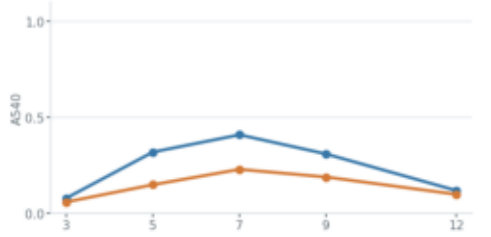
Amylase signal across physicochemical conditions

DNS endpoint A540 · Commercial fungal and salivary amylase

MALTOSE CALIBRATION REFIT
A540 = 0.81833C − 0.01578
R^2 = 0.99070

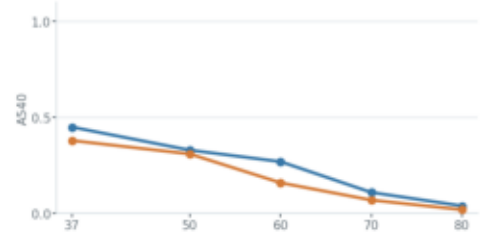
Commercial fungal amylase Salivary amylase Shared scale: 0–1.1 A540

pH



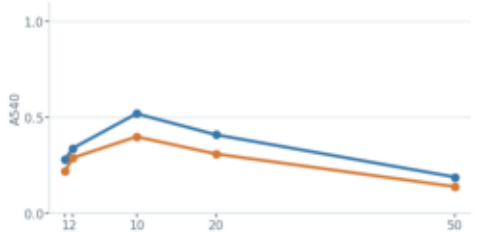
Highest at 7: fungal 0.41; salivary 0.23

Temperature



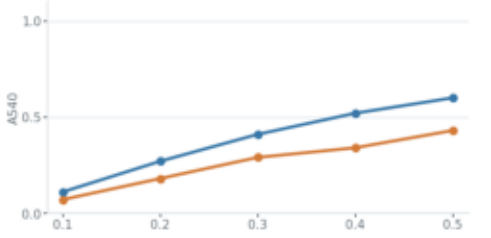
Highest at 37: fungal 0.45; salivary 0.38

$CaCl_2$



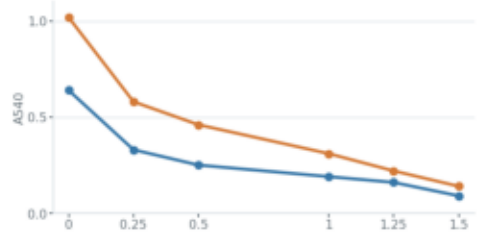
Highest at 10: fungal 0.52; salivary 0.40

Starch



Rises to 0.60 / 0.43 at 0.5%; no plateau

Ethanol



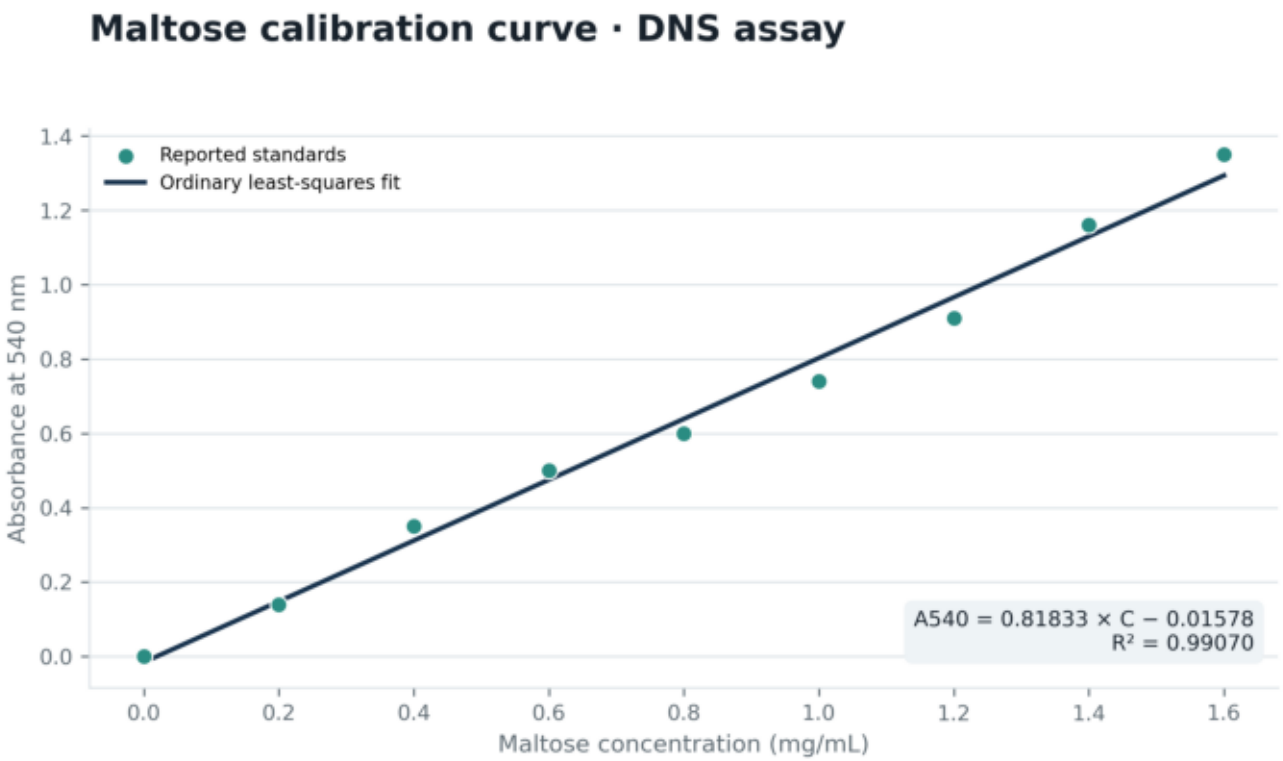
Falls to 0.09 / 0.14 by 1.5%

One source-reported reading per condition; no replicates or error bars. Lines are visual guides, not fitted models.

$CaCl_2$ and ethanol final concentrations are unclear. DNS results are endpoint absorbance, not normalized enzyme activity.

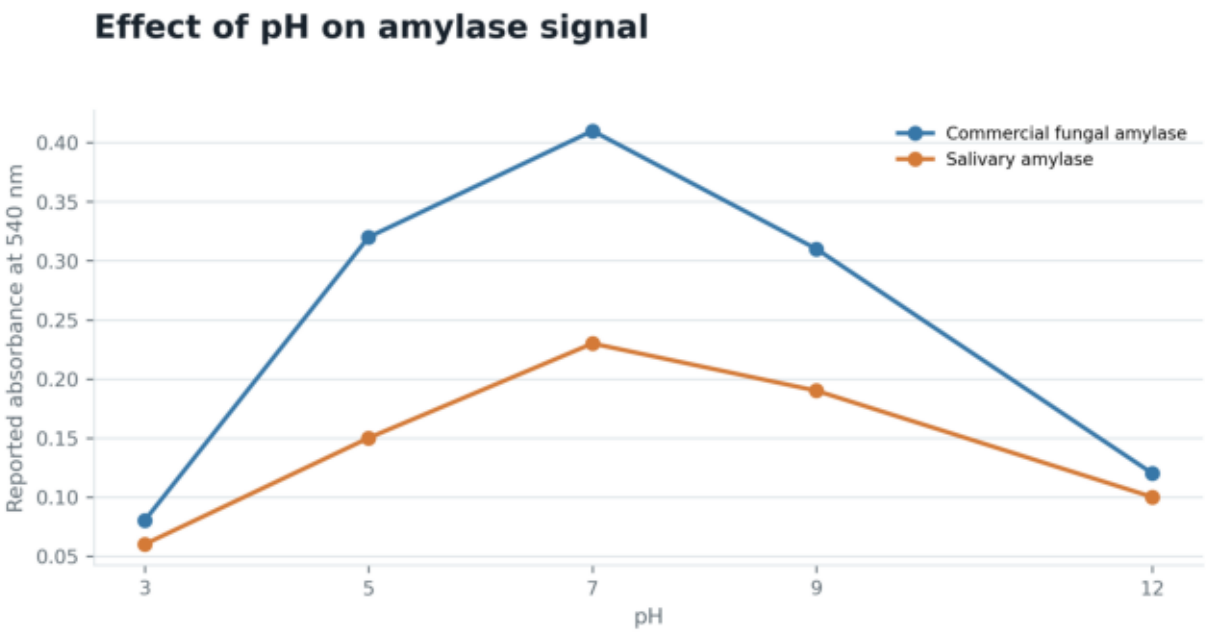
Source report: calibration pp. 8–9; assay tests pp. 13–32

Maltose calibration

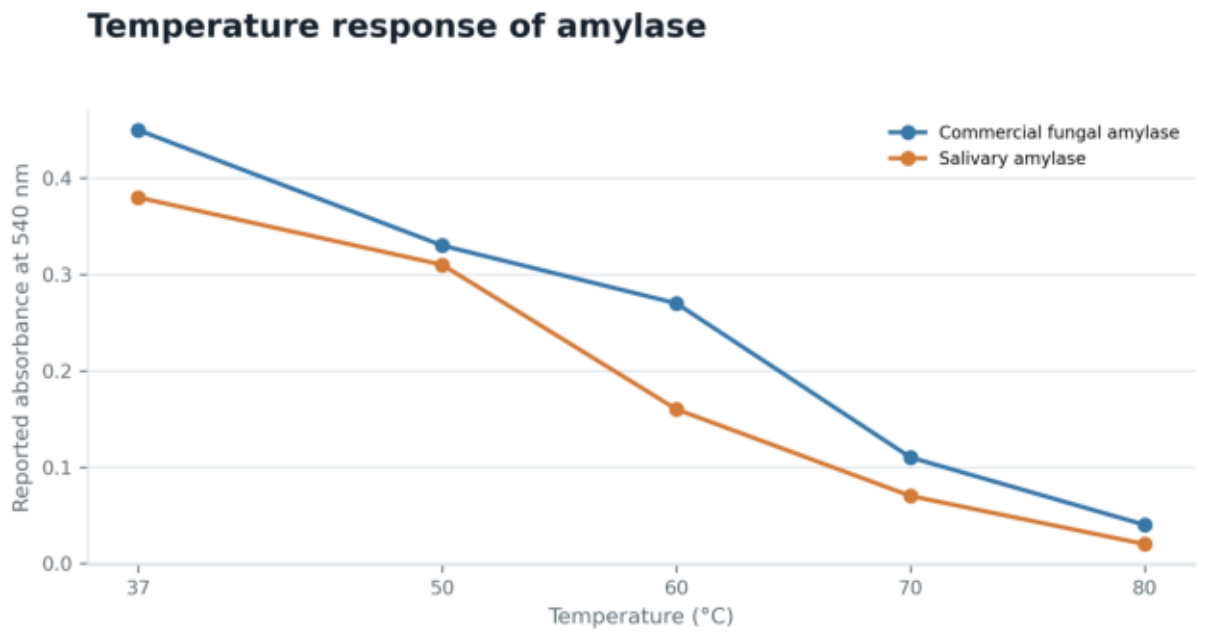


Fit recalculated from the nine tabulated standards; the source text contains an intercept-sign inconsistency.

pH and temperature responses

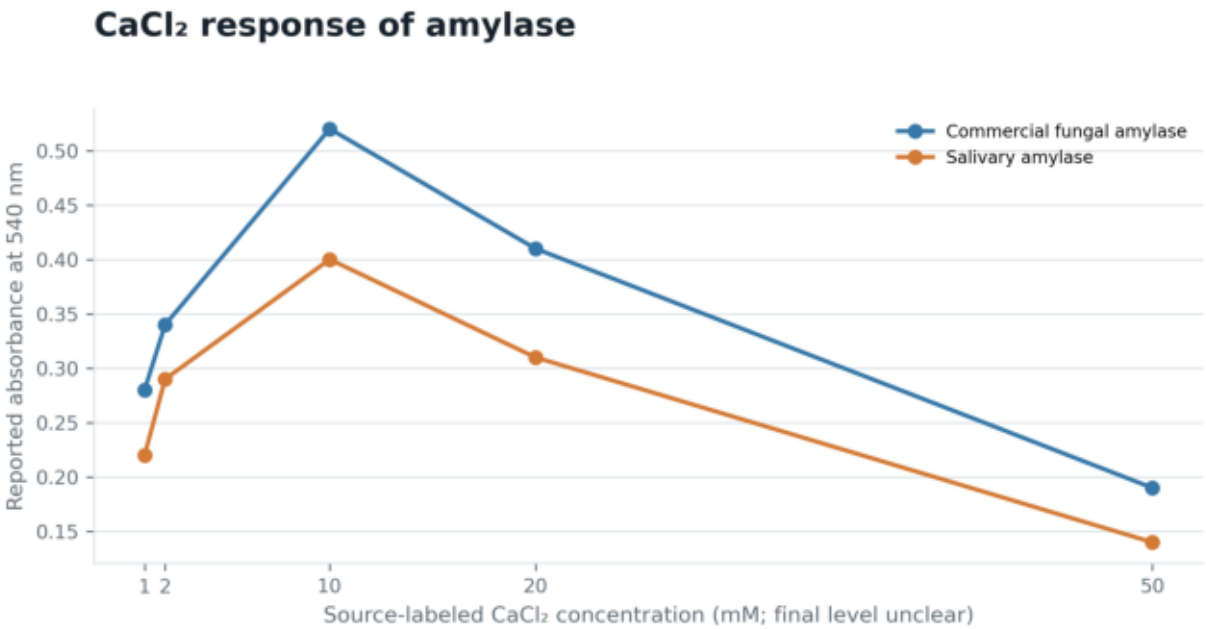


One source-reported reading per condition; connecting lines are visual guides, not fitted models.

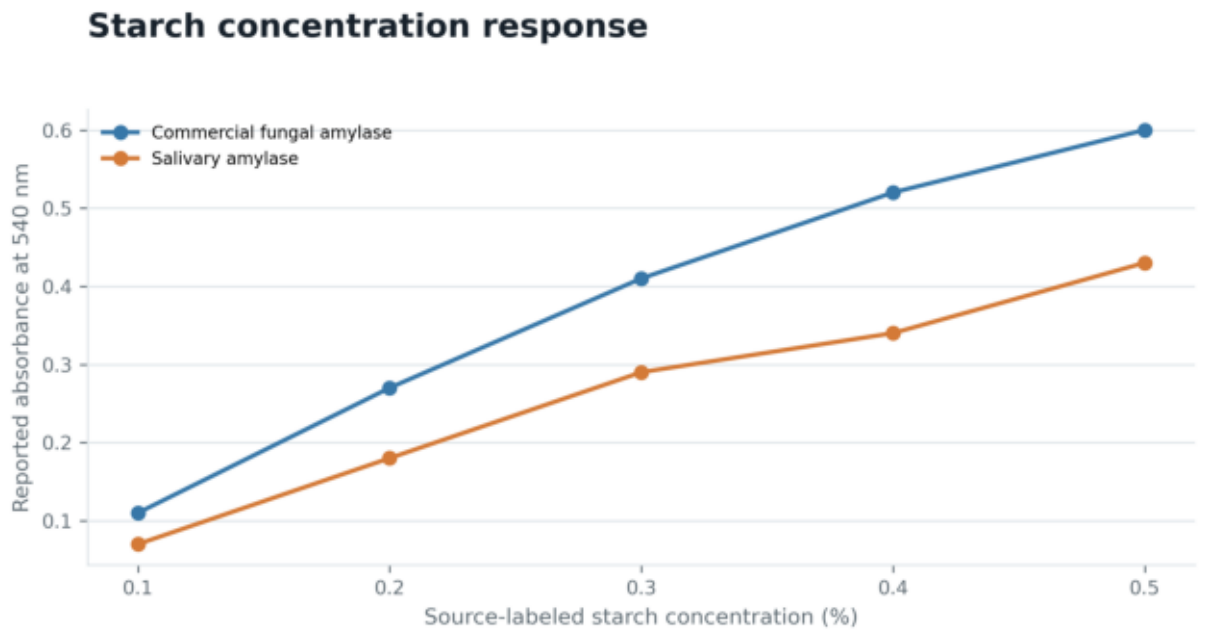


One source-reported reading per condition; connecting lines are visual guides, not fitted models.

Calcium and substrate responses

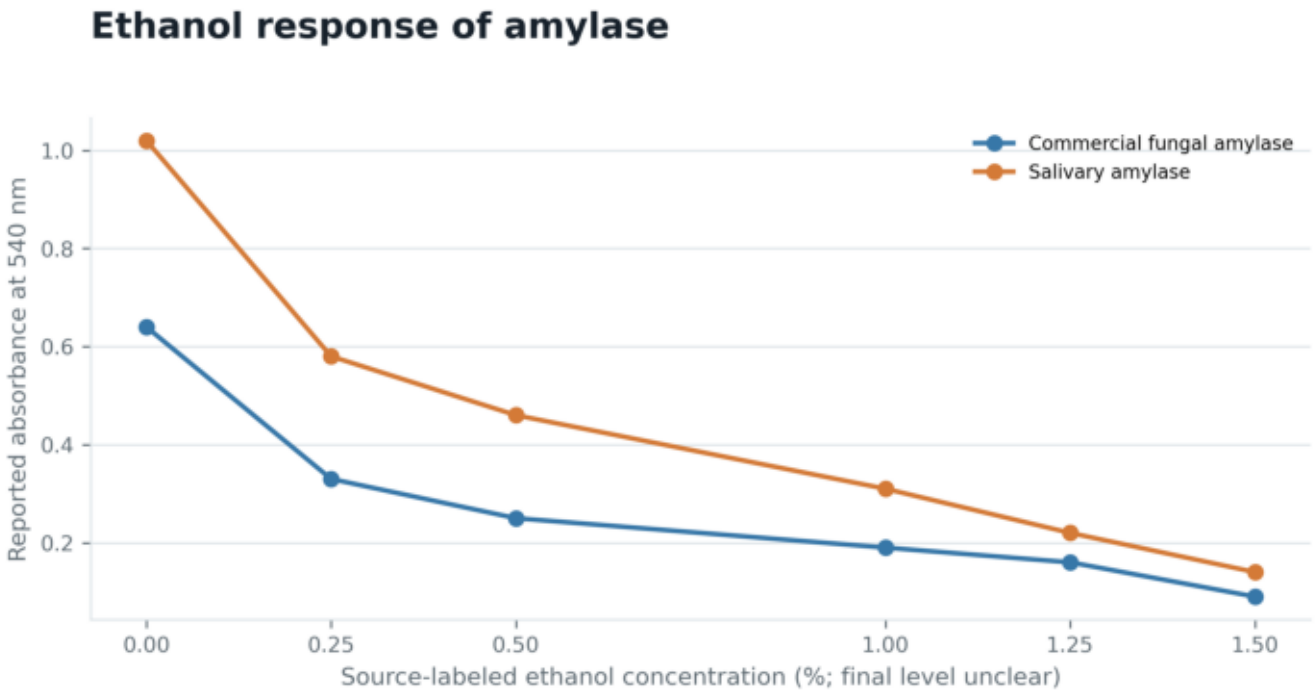


One source-reported reading per condition; connecting lines are visual guides, not fitted models.



One source-reported reading per condition; connecting lines are visual guides, not fitted models.

Ethanol response



One source-reported reading per condition; connecting lines are visual guides, not fitted models.

Interpretation and next steps

What the measurements support

Within the supplied one-reading-per-condition data, both preparations show their highest pH signal at 7 and highest temperature signal at 37°C. Both peak at the source-labeled 10 mM $CaCl_2$ condition. The starch response increases through 0.5%, while ethanol-associated absorbance decreases through 1.5%.

What they do not establish

There are no replicate values or uncertainty estimates, so the report does not support significance testing. The starch series does not demonstrate a plateau; Km and Vmax cannot be inferred. Different enzyme preparations are not normalized to protein or activity units.

Assay and reporting caveats

DNS responds to reducing sugars, so calibration output is maltose-equivalent. Buffer chemistry changes across the pH series; final $CaCl_2$ and ethanol concentrations are ambiguous. The source calibration text contains opposite intercept signs, and some ethanol concentrations do not match the refitted equation within rounding.

Recommended follow-up

Repeat with independent replicates, matched blanks and controls, recorded final reaction concentrations, and normalized enzyme input. Preserve raw spectrophotometer exports and refit the standard curve from replicate standards.